

Chapter X

Recurrent Iterated Function Systems

1 Fractal Systems

The goal of this chapter is to describe some general systems that can be used to construct deterministic fractals. We are concerned with the invention of mathematical machines for designing and controlling fractals.

We use the name “recurrent iterated function systems” loosely to encompass the systems introduced in this chapter. Examples include “IFS’s” in which the application of transformations to points depends on the “addresses” of the points. Other examples include “IFS’s with probabilities” which have invariant measures that can be computed by means of algorithms of the “Chaos Game” type, wherein, instead of each transformation being applied with a fixed probability, there are different probabilities that depend on which transformation was applied previously. A unified framework for presenting IFS, recurrent IFS, and future developments in the theory of deterministic fractals is provided by the idea of *fractal systems*.

In this section we introduce fractal systems, their associated fractals, and the objects or models which they can be used to approximate. We do this in such a way as to help the reader understand how to invent a fractal system appropriate to the type of model she or he is describing, without having first to develop a new theory from the bottom up. The ideas are loosely stated in order to convey most simply the underlying principles, which seem to be of general applicability and to be very powerful. From this author’s experience, it seems that any framework for a fractal system, once decided upon, can be readily fleshed out with the appropriate definitions and logical infrastructure to provide a full and clean theory.

Roughly speaking, a fractal system consists of a means for generating deterministic fractals, associated with an underlying space X , and a set or space of objects or

models, $\mathbb{Y}$, which these fractals can be used to approximate. Throughout this chapter we use $\square$ to denote a closed bounded subset of $\mathbb{R}^2$ such as $\{(x, y) \in \mathbb{R}^2 : 0 \leq x, y \leq 1\}$.

Already in this book we have looked at several very different fractal systems, as described in the following examples.

Example (i). One example of a fractal system is defined by IFS's of contractive affine transformations acting on $\mathbb{X} = \square \subset \mathbb{R}^2$: the underlying space is $\square$; the fractals of the system are attractors of all of the IFS's; and the space of models, $\mathbb{Y}$, which these fractals can be used to approximate, consists of all (compact) subsets of $\square$, namely $\mathbb{Y} = \mathcal{H}(\mathbb{X})$. In this example the models may represent monochrome pictures, with the sets representing the black parts of the pictures, and their complements representing the (background) white parts. The fractal system itself is a device for approximating the models by means of the fractals that it generates.

Example (ii). Another example of a fractal system is defined by the IFS's of contractive affine transformations on $\square \subset \mathbb{R}^2$, with probabilities. The fractals of this system are the attractors of the associated Markov operators, as described in Chapter IX. They lie in the space $\mathbb{Y} = \mathcal{P}(\square)$, the set of all normalized Borel measures on $\square \subset \mathbb{R}^2$. Here the underlying space $\mathbb{X}$ is $\square \subset \mathbb{R}^2$; its points do not belong to the fractals of the system because they are measures, not sets of points. In this example the models may represent grayscale pictures, with the amount of light reflected by a subset of the picture represented by the measure of that subset. Again, the fractal system provides a means for approximating the models, using the fractals that it generates.

Example (iii). A third example of a fractal system involves the space $\mathbb{X} = [0, 1] \subset \mathbb{R}$, with $\mathbb{Y} = C[0, 1]$, the continuous real valued functions on $[0, 1]$. In this system the fractals are the fractal interpolation functions on $[0, 1]$ that can be generated by affine transformations, as described in Chapter VI; and the models may represent time series data or stock market prices.

Further examples of fractal systems are introduced in this chapter, and involve recurrent iterated function systems. Yet others, involving local iterated function systems and appropriate to the modelling of real world images, are described in [BH].¹ All of these fractal systems use the same basic ingredients, which we describe next.

- (1) *An underlying metric space $(\mathbb{X}, d)$ needed to define the fractals and the models in the system.* For example, $\mathbb{X}$ may be $\mathbb{R}^2$, $\mathbb{R}^3$, or a subset of one of these spaces, such as $\square \subset \mathbb{R}^2$. Typically $(\mathbb{X}, d)$ is complete, and bounded subsets of it are compact.
- (2) *A space of models $\mathbb{Y} = \mathbb{Y}(\mathbb{X})$.* Each point of $\mathbb{Y}$ is a model, and models are defined with the aid of the space $\mathbb{X}$. The fractals generated by the fractal system

¹ M. F. Barnsley and L. P. Hurd, *Fractal Image Compression*, (1993), A. K. Peters, Wellesley, Massachusetts.

also belong to $\mathbb{Y}$. Examples of $\mathbb{Y}$ include spaces of sets, function spaces, and measure spaces. We also need a metric h on the space $\mathbb{Y}$, such that $(\mathbb{Y}, h)$ is a complete metric space. Examples of $(\mathbb{Y}, h)$ include $(\mathcal{H}(\mathbb{X}), d)$, the Hausdorff distance generated by d , $(C[0, 1], h(f, g) = \max\{|f(x) - g(x)| : x \in [0, 1]\})$, and $(\mathcal{P}(\mathbb{X}), d_h)$ where d_h is the Hutchinson distance between measures.

- (3) A contractive operator O which acts on the space $(\mathbb{Y}, h)$. That is, the operator O is such that there exists a real number s with $0 \leq s < 1$, and

$$h(O(\phi), O(\psi)) \leq s \cdot h(\phi, \psi) \text{ for all } \phi, \psi \in \mathbb{Y}.$$

Typically, the operator O is constructed with the aid of elementary contractive functions that act upon the underlying space $\mathbb{X}$.

For the examples of fractal systems, (i), (ii), and (iii) begun above, the operators O are set up as follows:

Example (i) (continued). $\mathbb{X} = \square \subset \mathbb{R}^2$, $w_i : \square \rightarrow \square$ is a contractive transformation for each $i = 1, 2, \dots, N$, $\mathbb{Y} = \mathcal{H}(\mathbb{X})$, and $O : \mathbb{Y} \rightarrow \mathbb{Y}$ is defined by

$$O(\mathbb{Y}) = W(\mathbb{Y}) = \bigcup \{w_i(x) : x \in \mathbb{Y}\}.$$

Remark that O is built up using contractive transformations acting on the underlying space.

Example (ii) (continued). $\mathbb{X} = \square \subset \mathbb{R}^2$, $\{\mathbb{X}; w_1, \dots, w_N; p_1, \dots, p_N\}$ are IFS's with probabilities, as in section 6 in Chapter IX, with the w_i 's as in (i) above, and $\mathbb{Y} = \mathcal{P}(\mathbb{X})$; we can then define a corresponding fractal system by taking O to be the associated Markov operator as defined in section 6 in Chapter IX; that is, $O : \mathbb{Y} \rightarrow \mathbb{Y}$ is defined by

$$O(v) = M(v) = p_1 v \circ w_1^{-1} + p_2 v \circ w_2^{-1} + \dots + p_N v \circ w_N^{-1}.$$

We know from the theory of Chapter IX that O is contractive in the Hutchinson metric. Again, remark that O is built up using contractive transformations, the w_i 's, acting on the underlying space.

Example (iii) (continued). In this example fractal interpolation functions, as described in Chapter VI, are used. $\mathbb{X} = [0, 1] \subset \mathbb{R}$, and $\mathbb{Y} = \mathcal{F} = C[0, 1]$. Then $O : \mathbb{Y} \rightarrow \mathbb{Y}$ is the operator $T : \mathcal{F} \rightarrow \mathcal{F}$ defined in the proof of Theorem 2.2 in Chapter VI. In the notation of section 2 in Chapter VI, this is, for $f \in \mathcal{F}$,

$$O(f) = (Tf)(x) = c_n l_n^{-1}(x) + d_n f(l_n^{-1}(x)) + f_n$$

for $x \in [x_{n-1}, x_n]$, for $n = 1, \dots, N$.

Then O is a contraction mapping with respect to the supremum distance on $C[0, 1]$. Again, notice how the operator O is built up from contractive transformations acting on the underlying space.

In practice, the contractive operator O is constructed from finite collections of contractive *affine* transformations; for example, affine transformations acting in two

dimensions. Furthermore, the coefficients of these constituent functions can be restricted to lie in finite sets, defined for example by rounding. Then O itself can be described by finite sets of discrete coefficients, the full set of which represents the “code” for the operator O .

The consequences of having the ingredients 1, 2, and 3 can be summarized in the following Theorems and Expectations:

Theorem 1.1 Existence of Attractors. *Since O is contractive, and the metric space $\mathbb{Y}$ is complete, there exists an unique attractor $\phi \in \mathbb{Y}$ such that*

$$O(\phi) = \phi$$

Proof See Theorem 6.1 in Chapter III.

Expectation 1.1 Fractal Character of Attractors. *We anticipate that ϕ is a fractal: that is, we expect that ϕ has a resolution-independent, infinitely-magnifiable character. This is because of the contractivity of the functions from which O is constructed: the whole invariant set is the same as a sum or union of contractions applied to it, and thus it is made of shrunk copies of (parts of) itself. Depending on the way in which the contractions act, the focus may be on contractivity of various spatial dimensions, and/or measure theoretic contractivity, and we expect that the attractor ϕ will inherit corresponding fractal characteristics.*

Theorem 1.2 Computation of Attractors. *To compute ϕ , we can use the fact that if $\psi \in \mathbb{Y}$ then the result of repeatedly applying O to ψ converges to the attractor ϕ ; that is,*

$$\lim_{n \rightarrow \infty} O^n(\psi) = \phi.$$

Moreover, if there exists a real constant C such that $h(\phi_1, \phi_2) < C$ for all $\phi_1, \phi_2 \in \mathbb{Y}$, then we have the error estimate

$$h(O^n(\psi), \phi) \leq s^n C.$$

Proof See Theorem 6.1 in Chapter III.

The latter equation tells us that the fixed point, or attractor, can be computed by algorithms of the “Photocopy Machine” type. The error estimate allows one to predict the number of iterations required to achieve a given accuracy.

Theorem 1.3 General Collage Theorem Estimate. *The distance between $\psi \in \mathbb{Y}$ and the attractor ϕ of O is bounded by the estimate*

$$h(\phi, \psi) \leq \frac{h(\psi, O(\psi))}{(1 - s)}.$$

Proof See Lemma 11.1 in Chapter III.

The set $O(\psi)$ is called a *collage* of ψ , while the distance $h(\psi, O(\psi))$ is called the corresponding *collage error*. The theorem says that if we wish to find an operator

O whose attractor ϕ is approximately ψ , then we have only to solve the problem of choosing O such that application of O to ψ does not change ψ very much, as discussed in Chapters III and IX.

This latter simple prescription is of central importance, yet easy to miss. We repeat it in slightly different terms. Suppose that we have a fractal system, defined by $\mathbb{Y}(\mathbb{X})$ and $\mathcal{O}$, where $\mathcal{O}$ is a set of operators O as above. We seek an operator $O \in \mathcal{O}$ whose attractor ϕ approximates $\psi \in \mathbb{Y}$. To do so, we adjust O so that the collage error $h(\psi, O(\psi))$ is as small as possible. Then O defines our fractal approximation ϕ . To store ϕ it suffices to store O . In the end, one thinks of the controllable family of operators $\mathcal{O}$ as the fractal system.

Whenever we have the ingredients 1, 2, and 3, we have a fractal system. A *fractal system* is a mechanism for finding fractal approximations ϕ for models $\psi \in \mathbb{Y}$. Examples (i), (ii), and (iii) above constitute interactively controllable systems for fractal modelling applied to diverse problems: in (i), sets of points are approximated by fractals made of sets of points; in (ii) measures are approximated by fractal measures; and in (iii), continuous functions are approximated by fractals that are graphs of continuous functions. In each case the parameters that control the problem are embedded in the operator O , which is selected and tailored to produce an apparatus for fractal approximation, a fractal system, appropriate to the modelling problem under consideration.

We note that fractal systems provide deterministic fractals rather than stochastic fractals. Once the operator O has been selected, its attractor, the fractal that it defines, has been fixed once and for all. There is only one fractal associated with each operator. These fractals are typically low information content members of a larger space, the space of models in which they lie; in much the same way as the rational numbers are finite information content members of the information rich space of all real numbers.

In the following sections we develop more elaborate fractal systems, based on recurrent IFS. Their introduction was begun in section 1 of this chapter. These generalize earlier systems in an elegant manner. They can be used to design machines to do elaborate fractal modelling, and illustrate well the design principles enunciated above.

2 Recurrent Iterated Function Systems

Recurrent iterated function systems were introduced in [BEH],² where more information can be found.

² M. Barnsley, J. Elton, D. Hardin, Recurrent Iterated Function Systems, Constructive Approximation (1989) 5, 3–31.

Table X.1. Example of a recurrent IFS code. Not all transitions are possible!

w_i	a_i	b_i	c_i	d_i	e_i	f_i	P_{i1}	P_{i2}	P_{i3}
1	0.5	0	0	0.5	0	0	0.3	0.7	0
2	0.5	0	0	0.5	0	128	0	0.6	0.4
3	0.5	0	0	0.5	128	128	0.5	0	0.5

Definition 2.1 A recurrent iterated function system consists of an IFS $\{\mathbb{X}; w_1, w_2, \dots, w_N\}$ together with a matrix $\{p_{ij} \in [0, 1] : i, j = 1, 2, \dots, N\}$, such that (i)

$$p_{i1} + p_{i2} + p_{i3} + \dots + p_{iN} = 1 \quad \text{for } i = 1, 2, \dots, N$$

and (ii) such that for any i and j , there exists a finite sequence of integers $k, l, \dots, m \in \{1, 2, \dots, N\}$ so that

$$p_{ik}p_{kl} \dots p_{mj} > 0$$

The transition probability for a certain discrete time Markov process is p_{ij} , which gives the chance, or probability, of moving from state i to state j , given that the process is in state i . Condition (i) says that the process is row stochastic: whichever state the system is in, a set of probabilities is available that sum to 1, and that describe the possible subsequent states to which the system can transition at the next step. Condition (ii) says that if the system is in state i then there is a finite probability of reaching state j in a finite number of steps, for any pair of integers $i, j \in \{1, 2, \dots, N\}$.

An IFS with probabilities provides a simple example of a recurrent IFS. The IFS with probabilities

$$\{\mathbb{X}; w_1, w_2, \dots, w_N; p_1, p_2, \dots, p_N\}$$

is in many ways the same as the recurrent IFS

$$\{\mathbb{X}; w_1, w_2, \dots, w_N; (p_{ij})\}$$

when $p_{ij} = p_j$ for all $i, j \in \{1, 2, \dots, N\}$.

We will say that a recurrent IFS is hyperbolic if the associated IFS is hyperbolic, that is, contractive with contractivity factor $0 \leq s < 1$. We restrict attention to hyperbolic recurrent IFS.

Examples of *codes* for recurrent IFS's are presented in Tables X.1 and X.2. In each case the space is $\mathbb{X} = \mathbb{R}^2$, the transformations are affine, and we use the usual notation

$$w_i \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_i \\ f_i \end{pmatrix}.$$

Figures X.266 and X.267 provide diagrams of the associated Markov processes. We say that the system is in *state* i if the last transformation to be applied was w_i . For the recurrent IFS in Table X.1 it is never possible to apply transformation

Table X.2. Example of a recurrent IFS code. All transitions can occur!

w_i	a_i	b_i	c_i	d_i	e_i	f_i	P_{i1}	P_{i2}	P_{i3}
1	0.5	0	0	0.5	0	0	0.3	0.6	0.1
2	0.5	0	0	0.5	0	128	0.1	0.5	0.4
3	0.5	0	0	0.5	128	128	0.4	0.4	0.2

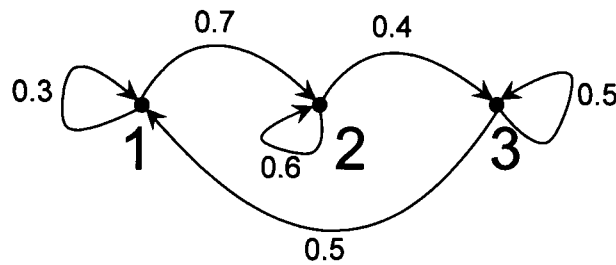


Figure X.266. Markov process for the recurrent IFS given in Table X.1. There is no path from state 1 to state 3, state 2 to state 1, or state 3 to state 2.

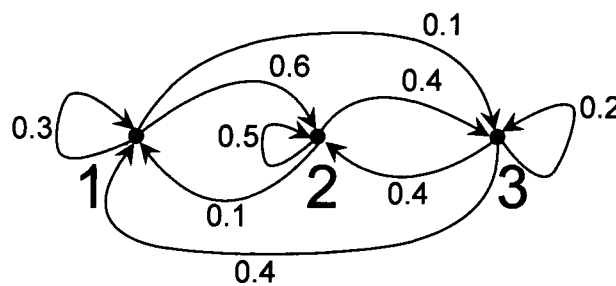


Figure X.267. Markov process for the recurrent IFS given in Table X.2. Paths exist between all possible states.

number 3 if the system is in state 1. It is also not possible to make a transition from state 2 to state 1. For the recurrent IFS in Table X.2, any transformation can follow any other one; that is, the system can transition from any state to any other state, in one step.

A recurrent IFS possesses a unique measure theoretic attractor, [BEH]. This is an invariant measure that can be computed by means of a generalized “Chaos Game” algorithm. It can also be computed using a wierd “Photocopy Machine.”

Example of the Chaos Game Algorithm for a Recurrent IFS. We present this algorithm for the case of the the recurrent IFS codes in Tables X.1 and X.2. The algorithm yields an orbit $\{(x_n, y_n) : n = 0, 1, 2, \dots\}$ that, with probability one, describes the unique invariant measure associated with the recurrent IFS. The relationship between the orbit and the measure is much the same as that described in Chapter IX for the case of standard IFS with probabilities.

- (1) Choose a starting point $(x_0, y_0) \in \mathbb{R}^2$. It is desirable that the starting point lies

as close as possible to the attractor. Clearly this lies in $[0, 256] \times [0, 256]$. Thus, we might choose $(x_0, y_0) = (0, 0)$.

- (2) Choose a initial state $s_0 = \{1, 2, 3\}$. Any other state in the set $\{1, 2, 3\}$ could be used. The Markov process associated with the transition matrix (p_{ij}) possesses a unique stationary vector (m_1, m_2, m_3) with $m_i > 0$, such that

$$\sum_{i=1}^3 m_i p_{ij} = m_j, \quad j = 1, 2, 3.$$

If one knows this vector, then it is a good idea to choose the initial state to be the one corresponding to the largest m_j .

- (3) Select $s_1 \in \{1, 2, 3\}$ with probability $p_{s_0 j}$ associated to the choice $s_1 = j$.
(4) Compute $(x_1, y_1) = w_{s_1}(x_0, y_0)$.
(5) Select $s_2 \in \{1, 2, 3\}$ with probability $p_{s_1 j}$ associated to the choice $s_2 = j$.
(6) Compute $(x_2, y_2) = w_{s_2}(x_1, y_1)$.

...

- (2n + 1) Select $s_n \in \{1, 2, 3\}$ with probability $p_{s_{n-1} j}$ associated to the choice $s_n = j$.

- (2n + 2) Compute $(x_n, y_n) = w_{s_n}(x_{n-1}, y_{n-1})$.

...

The result of the above calculation is supposed to be a long orbit

$$Q(\text{numits}) = \{(x_n, y_n) : n = 1, 2, \dots, \text{numits}\}.$$

(With probability one) the value of the invariant measure $\mu \in \mathcal{P}([0, 256] \times [0, 256])$ of the recurrent IFS is given by the formula

$$\lim_{\text{numits} \rightarrow \infty} \mu(S) = \frac{\text{number of points in } \{S \cap Q(\text{numits})\}}{\text{numits}}$$

for any measureable subset $S \subset \mathbb{R}^2$.

In Figure X.268 we illustrate this algorithm for the case of the recurrent IFS in Table X.1. In Figure X.269 we give the results of a computation, showing a long orbit. The invariant measure can be pictured by observing the density of the points.

In Figure X.270 we illustrate this algorithm for the case of the recurrent IFS in Table X.2. In Figure X.271 we give the results of a computation, showing a long orbit. Again, one gets some idea of the invariant measure by looking at the density of the points.

In Figure X.272 we show the structure and the attractor for the recurrent IFS in Table X.3. This involves four transformations which take $\square$ into itself. The range of w_i is the quadrant of $\square$ labeled i , so that the transformations provide a just-touching collage of the square. The transition probability p_{24} is zero, which means that no points on the attractor lie in the subquadrant labeled 24.

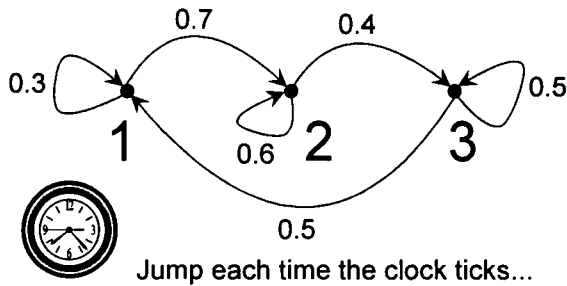
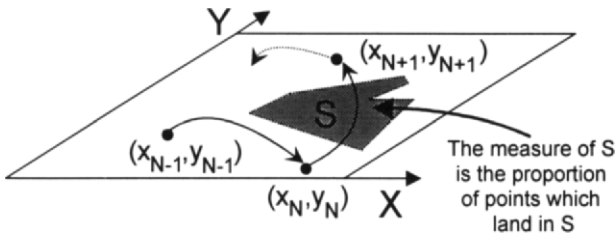


Figure X.268. Chaos Game for a recurrent IFS in which not all transitions are possible. The recurrent IFS code is given in Table X.1.

IN STATE 1	IN STATE 2	IN STATE 3
Jump to State 1 with Prob 0.3	Jump to State 2 with Prob 0.6	Jump to State 3 with Prob 0.5
Jump to State 2 with Prob 0.7	Jump to State 3 with Prob 0.4	Jump to State 1 with Prob 0.5



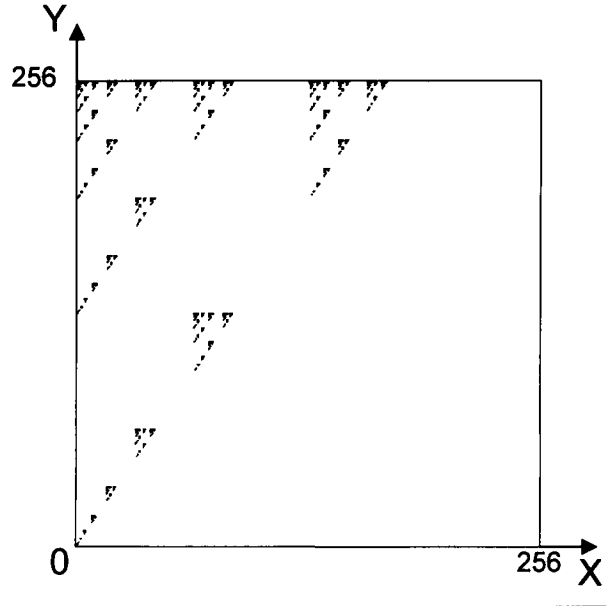
In the (x, y) plane, when the system jumps to State j , apply the map w_j to the current point (x_N, y_N) to obtain $(x_{N+1}, y_{N+1}) = w_j(x_N, y_N)$.

Thus produce a dancing point.

Figure X.273 shows the attractor for the maps in Table X.3, for six different sets of transition probabilities. These sets of transition probabilities used were:

- (1) $p_{11} = 0.3, p_{12} = 0.5, p_{13} = 0.1, p_{14} = 0.1,$
 $p_{21} = 0.1, p_{22} = 0.5, p_{23} = 0.4, p_{24} = 0.0,$
 $p_{31} = 0.3, p_{32} = 0.3, p_{33} = 0.2, p_{34} = 0.2,$
 $p_{41} = 0.25, p_{42} = 0.25, p_{43} = 0.25, p_{44} = 0.25;$
- (2) $p_{11} = 0.3, p_{12} = 0.5, p_{13} = 0.1, p_{14} = 0.1,$
 $p_{21} = 0.1, p_{22} = 0.5, p_{23} = 0.4, p_{24} = 0.0,$
 $p_{31} = 0.3, p_{32} = 0.3, p_{33} = 0.2, p_{34} = 0.2,$
 $p_{41} = 0.25, p_{42} = 0.25, p_{43} = 0.5, p_{44} = 0.0;$
- (3) $p_{11} = 0.3, p_{12} = 0.5, p_{13} = 0.1, p_{14} = 0.1,$
 $p_{21} = 0.1, p_{22} = 0.5, p_{23} = 0.4, p_{24} = 0.0,$
 $p_{31} = 0.0, p_{32} = 0.3, p_{33} = 0.5, p_{34} = 0.2,$
 $p_{41} = 0.0, p_{42} = 0.5, p_{43} = 0.5, p_{44} = 0.0.$
- (4) $p_{11} = 0.0, p_{12} = 0.5, p_{13} = 0.2, p_{14} = 0.3,$
 $p_{21} = 0.1, p_{22} = 0.4, p_{23} = 0.4, p_{24} = 0.1,$

Figure X.269. Some points on the orbit produced by the machinery in Figure X.268.

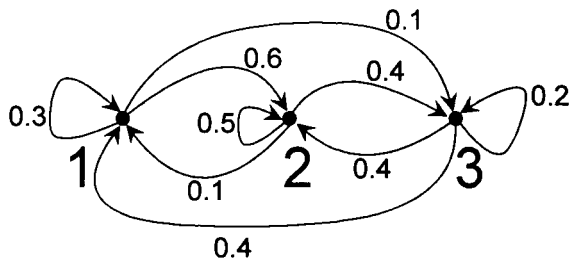


- $$p_{31} = 0.25, p_{32} = 0.25, p_{33} = 0.25, p_{34} = 0.25,$$
- $$p_{41} = 0.4, p_{42} = 0.4, p_{43} = 0.1, p_{44} = 0.1.$$
- (5) $p_{11} = 0.25, p_{12} = 0.25, p_{13} = 0.25, p_{14} = 0.25,$
 $p_{21} = 0.25, p_{22} = 0.25, p_{23} = 0.25, p_{24} = 0.25,$
 $p_{31} = 0.25, p_{32} = 0.25, p_{33} = 0.25, p_{34} = 0.25,$
 $p_{41} = 0.25, p_{42} = 0.25, p_{43} = 0.25, p_{44} = 0.25.$
- (6) $p_{11} = 0.0, p_{12} = 0.34, p_{13} = 0.33, p_{14} = 0.33,$
 $p_{21} = 0.25, p_{22} = 0.25, p_{23} = 0.25, p_{24} = 0.25,$
 $p_{31} = 0.25, p_{32} = 0.25, p_{33} = 0.25, p_{34} = 0.25,$
 $p_{41} = 0.25, p_{42} = 0.25, p_{43} = 0.25, p_{44} = 0.25.$

Can you work out which set of the above transition probabilities corresponds to each of the images in Figure X.273?

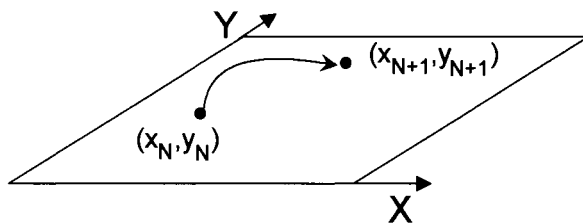
Example of the Photocopy Machine Algorithm for realizing the invariant measure of a recurrent IFS. We illustrate this algorithm for the case of the IFS code in Table X.1. It is illustrated in Figure X.274.

The mathematical machine in Figure X.274 corresponds to the recurrent IFS in Table X.1, wherein there are three affine transformations w_1 , w_2 , and w_3 , and wherein the transition probabilities p_{13} , p_{21} , and p_{32} are all zero. The states of the system are represented by the three copy units, one of which is labeled INPUT1/COPY1; the others are labeled INPUT2/COPY2 and INPUT3/COPY3. Each unit has two lens/filter components. The first copy unit is made of two lens/filter components; one of these components applies affine transformation w_1 to INPUT1,



Tic, Tic, Tic, Tic

IN STATE 1	IN STATE 2	IN STATE 3
Jump to State 1 with Prob 0.3	Jump to State 1 with Prob 0.1	Jump to State 1 with Prob 0.4
Jump to State 2 with Prob 0.6	Jump to State 2 with Prob 0.5	Jump to State 2 with Prob 0.4
Jump to State 3 with Prob 0.1	Jump to State 3 with Prob 0.4	Jump to State 3 with Prob 0.2



Apply map w_j when system jumps to State j .

Figure X.270. Random walk generator associated with the recurrent IFS in Table X.2.

attenuates the brightness of the image by factor p_{11} , and directs its output to COPY1; the other of these components applies affine transformation w_2 to the input image, attenuates the brightness of the image by a factor p_{12} , and directs its output to COPY2. The second copy unit has two lens/filter components; the first applies affine transformation w_2 to the input image INPUT2, attenuates the brightness of the image by factor p_{22} , and directs its output to COPY2; the other of these components applies affine transformation w_3 to the input image INPUT2, attenuates the brightness of the image by a factor p_{23} , and directs its output to COPY3. The third copy unit has two lens/filter components; the first applies affine transformation w_3 to the input image INPUT3, attenuates the brightness of the image by factor p_{33} , and directs its output to COPY3; the other of these components applies affine transformation w_1 to the input image INPUT3, attenuates the brightness of the image by a factor p_{31} , and directs its output to COPY1.

It is important to note that the three inputs and the three outputs are set up so that they all represent the same subset $\square$ of $\mathbb{R}^2$. The range of transformation w_1 lies in COPY1, the range of w_2 lies in COPY2, and the range of w_3 lies in COPY3. Thus,

Figure X.271. “Pic-
ture” of the point-set at-
tractor corresponding to
the fractal system repre-
sented in Figure X.270.

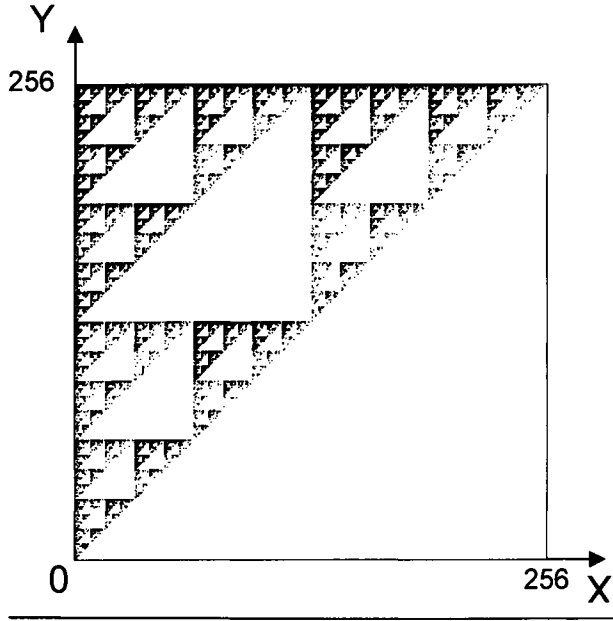


Table X.3. Recurrent IFS code. See Figure X.272.

w_i	a_i	b_i	c_i	d_i	e_i	f_i	P_{i1}	P_{i2}	P_{i3}	P_{i4}
1	0.5	0	0	0.5	0	0	0.3	0.5	0.1	0.1
2	0.5	0	0	0.5	0	128	0.1	0.5	0.4	0.0
3	0.5	0	0	0.5	128	128	0.3	0.3	0.2	0.2
4	0.5	0	0	0.5	128	0	0.25	0.25	0.25	0.25

in the first copy unit transformation $w_2 : \square \rightarrow \square$ in fact takes INPUT1 to COPY2, whilst in the second copy unit it takes INPUT2 to COPY2.

The images that are projected onto the copy screens are superposed; places where images come in from several inputs add up. For example, the total brightness of the image in COPY1 is p_{11} times the total brightness of the image in INPUT1, plus p_{31} times the total brightness of the image in INPUT3.

Copy images are recorded on “photographic paper” and put back on their respective inputs, as indicated in the figure. The system is cycled numerous times until the output images cease to change; then symbolically

$$\begin{aligned} \text{OUTPUT1} &= p_{11} \cdot \text{OUTPUT1} + p_{31} \cdot \text{OUTPUT3} \\ \text{OUTPUT2} &= p_{12} \cdot \text{OUTPUT1} + p_{22} \cdot \text{OUTPUT2} \\ \text{OUTPUT3} &= p_{23} \cdot \text{OUTPUT2} + p_{33} \cdot \text{OUTPUT3}. \end{aligned}$$

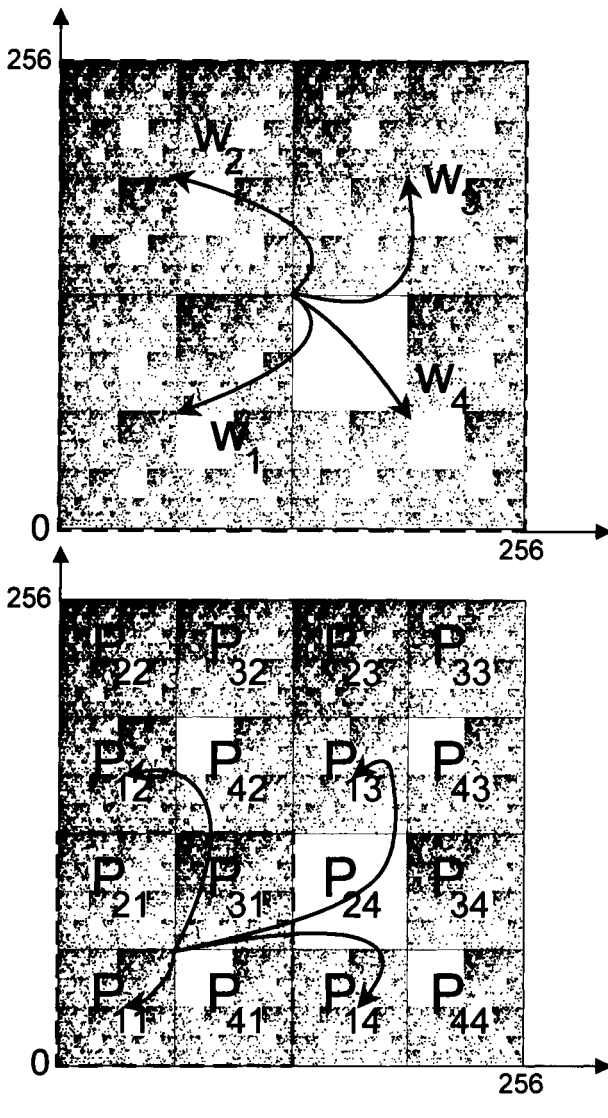


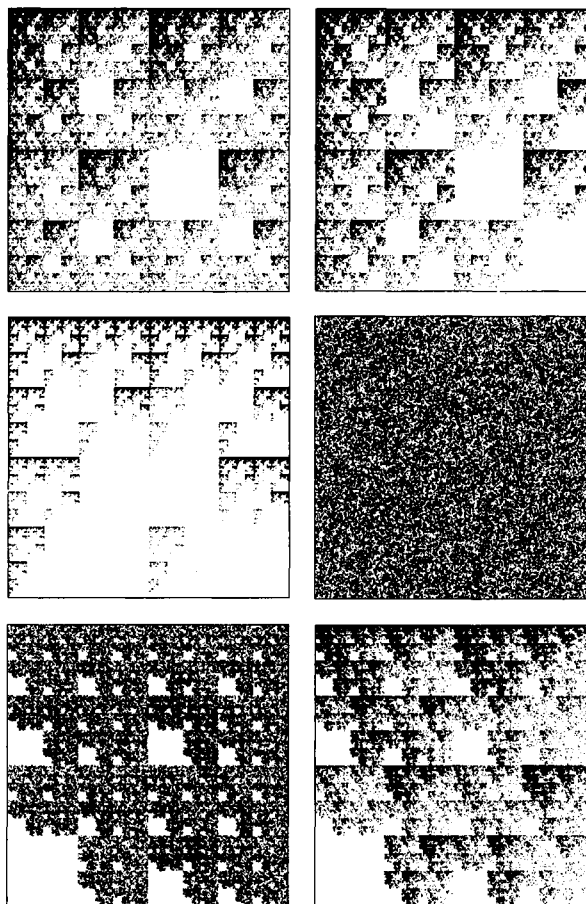
Figure X.272. Four affine transformations provide a just-touching collage of $\square$. A recurrent IFS (Table X.3) made of these transformations is such that $p_{24} = 0$. As a consequence, the attractor has no points in the square whose address is 24.

Finally, the images OUTPUT1, OUTPUT2 and OUTPUT3 are superposed. Symbolically,

$$\text{ATTRACTOR} = \text{OUTPUT1} + \text{OUTPUT2} + \text{OUTPUT3}.$$

Color Plate 37 illustrates a measure theoretic attractor of a related Photocopy Machine. Color Plate 40 illustrates another way of approaching the computation of the invariant measure of a recurrent IFS, using cameras and screens.

Figure X.273. All of these attractors correspond to different choices for the transition probabilities in Table X.3.



3 Collage Theorem for Recurrent IFS

In this section we are concerned with the construction of fractal systems, based on recurrent IFS, with an emphasis on point-set attractors, that is, attractors that are sets of points rather than measures. Our goal is to describe how a collage theorem and associated machinery can be set up to enable fractal modelling using the point-set attractors of certain recurrent IFS.

A recurrent IFS can be described in terms of a Markov chain which acts on a code space built from the symbols $\{1, 2, \dots, N\}$, as illustrated in Figures X.268 and X.270. The numbers $p_{ij} \geq 0$, $\sum_{j=1}^N p_{ij} = 1$, give the probabilities of transfer among the symbols. One can imagine a particle moving from symbol to symbol following the discrete time Markov process. The process is defined, strictly speaking, to be *recurrent* if there is a finite probability of being able to move, on the directed graph,

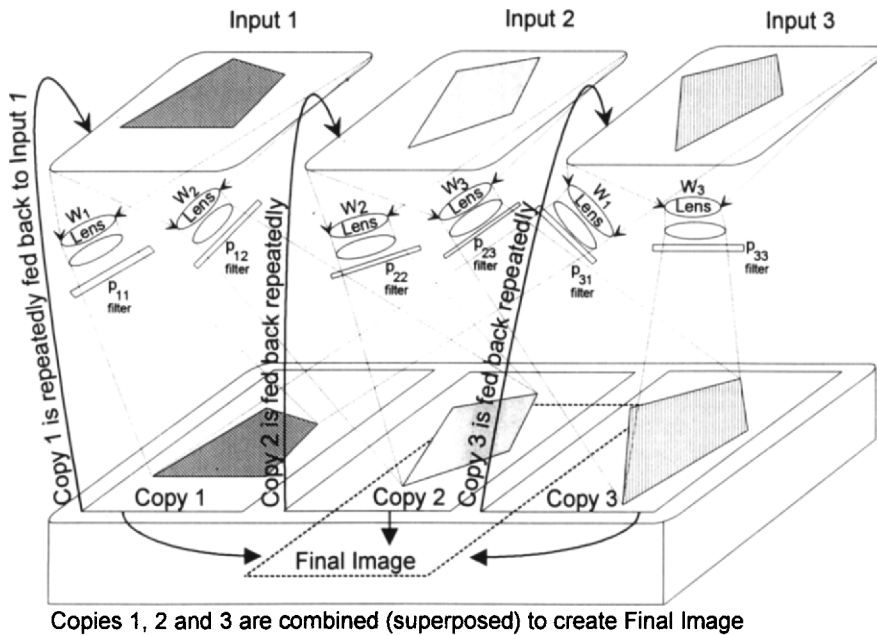


Figure X.274. Photocopy Machine for realizing the invariant measure of the recurrent IFS in Table X.1.

from any given symbol to any given symbol. A good source of information on Markov chains is [Feller, Ch. 15].³

We are concerned with the *hyperbolic* case, namely,

$$d(w_i(x), w_i(y)) \leq sd(x, y) \quad \forall i, \forall x, y \in \mathbb{X}$$

where d is the distance function on $\mathbb{X}$ and $0 \leq s < 1$. In this case, we know that there exists a unique attractive invariant probability measure μ , which describes the random walk on $\mathbb{X}$ produced for example by the “Chaos Game” algorithm described in the previous section. Our focus in this section is on the structure of the support of μ , which we call $A \in \mathbb{X}$, the attractor of the recurrent IFS. This depends only on which p_{ij} are nonzero, the *connection structure* of the Markov chain, and otherwise not on the values of the p_{ij} ’s.

We begin by recalling how the analysis proceeds in the case of IFS theory; this provides the simplest framework for explaining the recurrent case. Let $(\mathbb{X}, d)$ denote a compact metric space $\mathbb{X}$, with distance function d . Let $\mathcal{H}$ denote the set of all nonempty compact subsets of $\mathbb{X}$.

Definition 3.1 $d(x, B) = \min_{y \in B} d(x, y) \quad \forall x \in \mathbb{X}, \forall B \in \mathcal{H}$. Note that

$$B \subset C \Rightarrow d(x, C) \leq d(x, B). \quad (1)$$

³ W. Feller, An Introduction to Probability Theory and its Applications (Wiley, London), 1957.

Definition 3.2 $d(A, B) = \max_{x \in A} d(x, B) \quad \forall A, B \in \mathcal{H}$.

Note that this set distance is not symmetric. It has the following properties

- (i) $B \subset C \Rightarrow d(A, C) \leq d(A, B)$. This follows at once from (1).
- (ii) $d(A \cup B, C) = d(A, C) \vee d(B, C)$, where $x \vee y = \max\{x, y\}$. This follows from the observation that

$$d(A \cup B, C) = \max_{x \in A \cup B} d(x, C) = \max_{x \in A} d(x, C) \vee \max_{x \in B} d(x, C).$$

Definition 3.3 For all $A, B \in \mathcal{H}$, the Hausdorff distance is defined by

$$h(A, B) = d(A, B) \vee d(B, A).$$

We recall that $(\mathcal{H}, h)$ is a compact metric space.

Lemma 3.1 For all $A, B, C, D \in \mathcal{H}$,

$$h(A \cup B, C \cup D) \leq h(A, C) \vee h(B, D).$$

Proof

$$\begin{aligned} d(A \cup B, C \cup D) &= d(A, C \cup D) \vee d(B, C \cup D) \quad \text{by (ii)} \\ &\leq d(A, C) \vee d(B, D) \quad \text{by (i)} \\ &\leq h(A, C) \vee h(B, D). \end{aligned}$$

The same argument yields $d(C \cup D, A \cup B) \leq d(C, A) \vee d(D, B) \leq h(A, C) \vee h(B, D)$.

Let $\{\mathbb{X}; w_j, j = 1, 2, \dots, N\}$ be a hyperbolic IFS, with $d(w_j x, w_j y) \leq s d(x, y) \quad \forall x, y \in \mathbb{X}$, and $0 \leq s < 1$. Define

$$W : \mathcal{H} \rightarrow \mathcal{H}$$

by

$$W(A) = \bigcup_{j=1}^N w_j(A).$$

Theorem 3.1 $W : \mathcal{H} \rightarrow \mathcal{H}$ is a contraction, with contractivity factor s , with respect to the Hausdorff metric; that is,

$$h(W(A), W(B)) \leq s h(A, B) \quad \forall A, B \in \mathcal{H}.$$

Proof For any $A, B \in \mathcal{H}$,

$$\begin{aligned}
 h(W(A), W(B)) &= h\left(\bigcup_{j=1}^N w_j(A), \bigcup_{j=1}^N w_j(B)\right) \\
 &\leq \bigvee_{j=1}^N h(w_j(A), w_j(B)) \quad (\text{by Lemma 3.1}) \\
 &= \bigvee_{j=1}^N \{d(w_j(A), w_j(B)) \vee d(w_j(B), w_j(A))\} \\
 &\leq \bigvee_{j=1}^N \{sd(A, B) \vee sd(B, A)\} \\
 &= sh(A, B)
 \end{aligned}$$

We leave it as an exercise for the reader to show that the following extension of the above result is true. We will need this fact later on. If $(\mathbb{X}_1, d_1)$ and $(\mathbb{X}_2, d_2)$ are metric spaces, then $(\mathcal{H}_1, h_1)$ and $(\mathcal{H}_2, h_2)$ are the corresponding spaces of compact nonempty subsets, if

$$w_j : \mathbb{X}_1 \rightarrow \mathbb{X}_2 \quad \text{for } j = 1, 2, \dots, N,$$

obeys

$$d_2(w_j x, w_j y) \leq sd_1(x, y) \quad \forall x, y \in \mathbb{X}_1,$$

then

$$W : \mathcal{H}_1 \rightarrow \mathcal{H}_2$$

defined by

$$W(A) = \bigcup_{j=1}^N w_j(A)$$

obeys

$$h_2(W(A), W(B)) \leq sh_1(A, B).$$

Corollary 3.1 *There is a unique set $A \in \mathcal{H}$ such that $W(A) = A$.*

A is the attractor of the IFS.

Corollary 3.2 Collage Theorem for IFS. *If $B \in \mathcal{H}$ obeys*

$$h(B, W(B)) \leq \epsilon > 0$$

then

$$h(B, A) \leq \frac{\epsilon}{(1-s)},$$

where A denotes the attractor of the IFS.

Proof By the contraction mapping theorem

$$h(A, B) = h\left(B, \lim_{n \rightarrow \infty} W^{\circ n}(B)\right) = \lim_{n \rightarrow \infty} h(B, W^{\circ n}(B))$$

where $W^{\circ 0}(B) = W(B)$ and we inductively define

$$W^{\circ(n+1)}(B) = W(W^{\circ n}(B)), \quad n = 0, 1, 2, \dots$$

But, by the triangle inequality,

$$\begin{aligned} h(B, W^{\circ n}(B)) &\leq \sum_{m=1}^n h(W^{\circ(m-1)}(B), W^{\circ m}(B)) \\ &= \sum_{m=1}^n h(W^{\circ(m-1)}(B), W^{\circ(m-1)}(W(B))) \\ &\leq \sum_{m=1}^n s^{m-1} h(B, W(B)) \\ &\leq (1-s)^{-1} h(B, W(B)) \end{aligned}$$

We are now ready to extend the structure to the case of recurrent IFS. We actually make a generalization of the recurrent IFS structure to multiple spaces and set maps, suitable for the hyperbolic case where one is concerned with point-set topology issues. We are only concerned here with the connection structure of the chain. Let (X_j, d_j) be compact metric spaces, $j \in \{1, 2, \dots, N\}$. Let $(\mathcal{H}_j, h_j)$ denote the associated metric spaces of non-empty compact subsets which use the Hausdorff metrics. Let there be defined maps $W_{ij} : \mathcal{H}_j \rightarrow \mathcal{H}_i, \forall (i, j) \in I$ where I is some set of pairs of indices with the property that for each $i \in \{1, 2, \dots, N\}$ there is a $j \in \{1, 2, \dots, N\}$ with $(i, j) \in I$. That is, $I(i) = \{j | (i, j) \in I\} \neq \emptyset$ for each $i \in \{1, 2, \dots, N\}$. Furthermore let

$$h_i(W_{ij}(A), W_{ij}(B)) \leq s_{ij} h_j(A, B)$$

for some number $s_{ij}, \forall (i, j) \in I, \forall A, B \in \mathcal{H}_j$. By the remark following Theorem 3.1 such maps can be built up from point maps taking X_j to X_i .

Suppose $w_i : X \rightarrow X$ are contractive maps, where X is compact metric, and (p_{ij}) is row stochastic. Define $(X_j, d_j) = (X, d)$ for each j , and define $W_{ij}(S) = \{w_i(x) : x \in S\}, i, j = 1, \dots, N$. Let $I(i) = \{j : p_{ji} > 0\}$. This embeds us in the more general setup that we now study.

Let

$$\tilde{\mathcal{H}} = \mathcal{H}_1 \times \mathcal{H}_2 \times \mathcal{H}_3 \times \dots \times \mathcal{H}_N,$$

and endow $\tilde{\mathcal{H}}$ with the metric $\tilde{h}$ defined by

$$\tilde{h}((A_1, A_2, \dots, A_N), (B_1, B_2, \dots, B_N)) = \max \{h_j(A_j, B_j) | j = 1, 2, \dots, N\}.$$

Then it is readily demonstrated by the enthusiastic reader that $(\tilde{\mathcal{H}}, \tilde{h})$ is a compact metric space.

We think of $\tilde{\mathcal{H}}$ as consisting of a stack of clipped planes $\mathbb{X}_1, \mathbb{X}_2, \dots, \mathbb{X}_N$ with a point in $\tilde{\mathcal{H}}$ being the N -tuple of one image in each plane. This idea is illustrated in Color Plate 41. Define

$$\tilde{W} : \tilde{\mathcal{H}} \rightarrow \tilde{\mathcal{H}}$$

by

$$\tilde{W}(A_1, A_2, \dots, A_N) = \left(\bigcup_{j \in I(1)} w_{1j}(A_j), \bigcup_{j \in I(2)} w_{2j}(A_j), \dots, \bigcup_{j \in I(N)} w_{Nj}(A_j) \right)$$

For example, in the case $N = 2$ such a mapping might be symbolized

$$\tilde{W} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} \emptyset & W_{12} \\ W_{21} & W_{22} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} W_{12}(A_2) \\ W_{21}(A_1) \cup W_{22}(A_2) \end{pmatrix}$$

We have now arrived at our goal; we are able to characterize $\tilde{W}$ as a contraction mapping on $\tilde{\mathcal{H}}$; we thus have all of the ingredients of a fractal system.

Theorem 3.2 $\tilde{W} : \tilde{\mathcal{H}} \rightarrow \tilde{\mathcal{H}}$ obeys

$$\tilde{h}(\tilde{W}(A), \tilde{W}(B)) \leq s \tilde{h}(A, B) \quad \forall A, B \in \tilde{\mathcal{H}},$$

where $s = \max\{s_{ij}, (i, j) \in I\}$.

Proof To keep the notation succinct we suppose

$$\tilde{W} = \begin{pmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{pmatrix}$$

Then if $A = (A_1, A_2)$ and $B = (B_1, B_2)$ we have

$$\begin{aligned} & \tilde{h}(\tilde{W}(A), \tilde{W}(B)) \\ &= \tilde{h}((W_{11}(A_1) \cup W_{12}(A_2), W_{21}(A_1) \cup W_{22}(A_2)), \\ & \quad (W_{11}(B_1) \cup W_{12}(B_2), W_{21}(B_1) \cup W_{22}(B_2))) \\ &= \max\{h_1(W_{11}(A_1) \cup W_{12}(A_2), W_{11}(B_1) \cup W_{12}(B_2)), \\ & \quad h_2(W_{21}(A_1) \cup W_{22}(A_2), W_{21}(B_1) \cup W_{22}(B_2))\} \\ &\leq \max\{h_1(W_{11}(A_1), W_{11}(B_1)) \vee h_1(W_{12}(A_2), W_{12}(B_2)), \\ & \quad h_2(W_{21}(A_1), W_{21}(B_1)) \vee h_2(W_{22}(A_2), W_{22}(B_2))\} \quad (\text{by Lemma 3.1}) \\ &\leq \max\{s_{11}h_1(A_1, B_1) \vee s_{12}h_2(A_2, B_2), s_{21}h_1(A_1, B_1) \vee s_{22}h_2(A_2, B_2)\} \\ &\leq s(h_1(A_1, B_1) \vee h_2(A_2, B_2)) = s \tilde{h}((A_1, A_2), (B_1, B_2)) \\ &= s \tilde{h}(A, B). \end{aligned}$$

The fact that W is a contraction mapping on $\tilde{\mathcal{H}}$ means that we have a fractal system as described in the first section of this chapter. Conditions 1, 2, and 3, of section 1 apply. In this case the underlying space is $\mathbb{X}$. Multiple copies of $\mathbb{X}$ are used to make up a space of models $\mathbb{Y} = \tilde{\mathcal{H}}$. (It may be that we only care about one “plane” in $\tilde{\mathcal{H}}$.) A contraction mapping $O = \tilde{W}$ on $\tilde{\mathcal{H}}$ is built up out of elementary

transformations acting on $\mathbb{X}$. Fixed points of such operators O provide the fractal approximations to the models. In the present case the specific structure is as follows.

Corollary 3.3 *When $s < 1$ there is a unique element*

$$A = (A_1, A_2, \dots, A_N) \in \tilde{\mathcal{H}}$$

such that

$$A_i = \bigcup_{j \in I(i)} W_{ij}(A_j) \quad \text{for } i = 1, 2, \dots, N.$$

That is to say, $\tilde{W}(A) = A$. A is the attractor of the recurrent IFS.

Corollary 3.4 A Collage Theorem for recurrent IFS. *If $B \in \tilde{\mathcal{H}}$ obeys*

$$\tilde{h}(B, \tilde{W}(B)) \leq \epsilon > 0$$

then

$$h(B, A) \leq \frac{\epsilon}{(1-s)},$$

where A denotes the attractor of the recurrent IFS.

To connect this with the original single-space point-map recurrent IFS, we have

Corollary 3.5 *Let $\{\mathbb{X}; w_i, (p_{ij}), i, j = 1, \dots, N\}$ be a recurrent IFS with $\mathbb{X}$ compact and the w_i 's uniform contractions. Let A be the support of the unique stationary measure μ mentioned in section 2. Then there exist unique compact sets $A_i \subset A$, $i = 1, \dots, N$ with $A = \bigcup_{i=1}^N A_i$ such that*

$$A_i = \bigcup_{j: p_{ji} > 0} w_i(A_j), \quad i = 1, \dots, N.$$

In terms of the random walk produced by the Chaos Game Algorithm, as illustrated in section 2, the A_i 's may be characterized as follows: for all x , $x \in A_i$ iff for every neighborhood G of x , for almost all trajectories $x_0, w_{i_1}(x_0), w_{i_2}(w_{i_1}(x_0)), \dots$, we have $i_n = i$ and $w_{i_n}(\dots(w_{i_1}x_0)\dots) \in G$ for infinitely many n . In other words, to "see" A_i , just look at the points along a trajectory that end in map w_i .

Let us use the above ideas to construct a simple machine (fractal system) for modelling some binary subsets of $\square$. The machine is illustrated in Figure X.275. It uses affine transformations $w_i: \square \rightarrow \square$, $i = 1, 2, 3$. Place a copy of the subset $T \subset \square$ on (1), and place *adjustable* subsets $U, V \subset \square$ on (2) and (3) respectively. Adjust the coefficients $a_i, b_i, c_i, d_i, e_i, f_i$ and the two subsets U and V , so that

$$T \simeq w_1(T) \cup w_2(U) \cup w_3(V)$$

$$U \simeq w_1(T) \cup w_2(U)$$

$$V \simeq w_1(T) \cup w_3(V)$$

Assuming that the system can be adjusted so that these equations hold approximately, while the transformations remain suitably contractive, we know that the attractor for the system, our fractal model, which will be represented by a vector of sets $(\hat{T}, \hat{U}, \hat{V})$, will be such that $\hat{T} \simeq T$. This simple machine generates an 18 parameter family of fractal models for subsets of $\square$.

A simple machine for modelling some binary subsets of $\square$.

$$w_i \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_i \\ f_i \end{pmatrix}$$

for $i = 1, 2, 3$

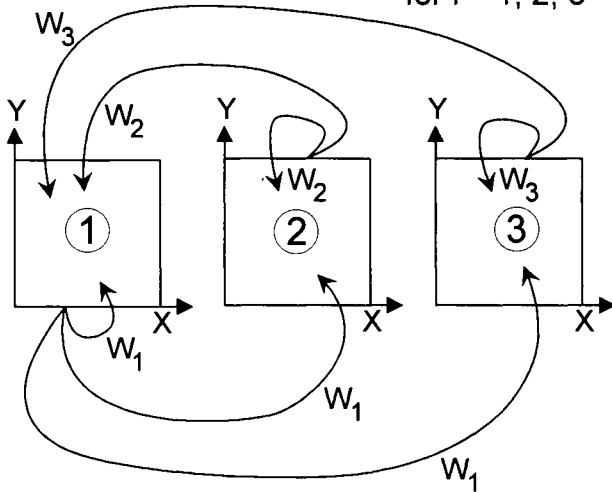


Figure X.275. Imagine that you have control knobs so that you can adjust the affine coefficients.

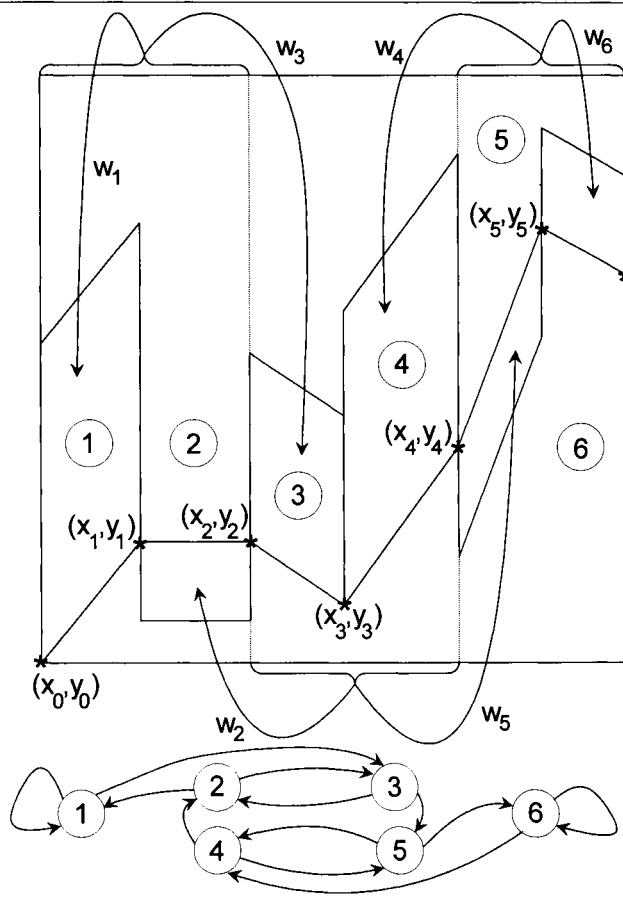
Another simple point-set fractal system is suggested by the diagram in Figure X.276. This type of machine generates recurrent fractal interpolation functions. The coefficients of the transformations are constrained to make sure that the resulting attractors are graphs of continuous functions, as follows:

$$\begin{array}{lll} w_1(x_0, y_0) = (x_0, y_0), & w_3(x_2, y_2) = (x_3, y_3), & w_5(x_2, y_2) = (x_4, y_4) \\ w_1(x_2, y_2) = (x_1, y_1), & w_3(x_0, y_0) = (x_2, y_2), & w_5(x_4, y_4) = (x_5, y_5) \\ w_2(x_2, y_2) = (x_1, y_1), & w_4(x_4, y_4) = (x_3, y_3), & w_6(x_4, y_4) = (x_5, y_5) \\ w_2(x_4, y_4) = (x_2, y_2), & w_4(x_6, y_6) = (x_4, y_4), & w_6(x_6, y_6) = (x_6, y_6). \end{array}$$

One example of a commercial fractal system, useful for educational purposes and for desktop publishing applications, is the *Desktop Fractal Design System*, [DFDS].⁴ This is based on simple recurrent IFS theory using affine transformations in two dimensions. This can be used to illustrate some aspects of modelling using point-set attractors of recurrent IFS built up of affine transformations acting in two dimensions.

⁴ The Desktop Fractal Design System, Version 2.0, for MacIntosh and IBM PC. Published by Academic Press, 1992.

Figure X.276. A fractal system constrained so that its attractors are graphs of continuous functions.



Consider the IFS for a fern, as introduced in Chapter III. We refer to this as Fern 1. Its IFS code is given in Table X.4 and it is shown in Figure X.277.

In Figure X.278 we illustrate how one can create the same picture of Fern 1 by using a duplicate copy of the space $\square$ with the same IFS acting on that space, and allowing a transformation to map from one copy of the space to the other.

Look closely at the structure of Fern 1. Notice how the main fronds alternate as we go up the main stem. As we saw in Chapter III, this same structure is repeated at all scales. If we zoom in on the tiniest frond, we will find that all the frondlets alternate up their respective stems.

Now suppose we modify our more complex recurrent system by adjusting the IFS for the first copy of the fern, that is we adjust the affine coefficients so that Fern 1 on the first copy of $\square$ becomes Fern 2, as illustrated in Figure X.279 and Table X.5.

Notice that the fronds in Fern 2 are opposed instead of alternating. Finally, in Figure X.280, we show the effect of the modifications on the attractor as a whole. The original copy of the fern now has opposed fronds, some of which are made of

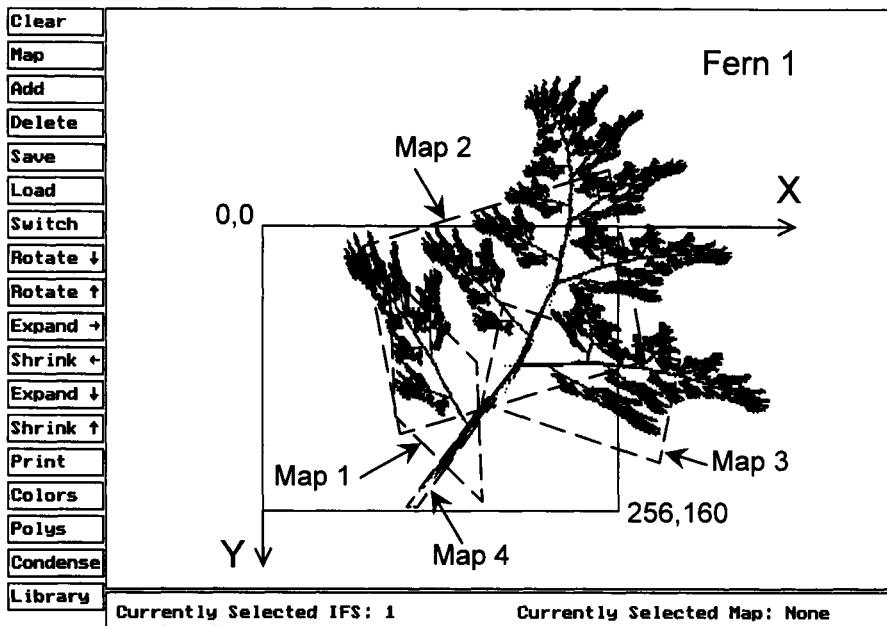


Figure X.277. A binary fern image is the point-set attractor for a single fractal system consisting of an IFS of affine maps acting in $\mathbb{R}^2$. It is closely related to more complex fractals based on recurrent IFS theory.

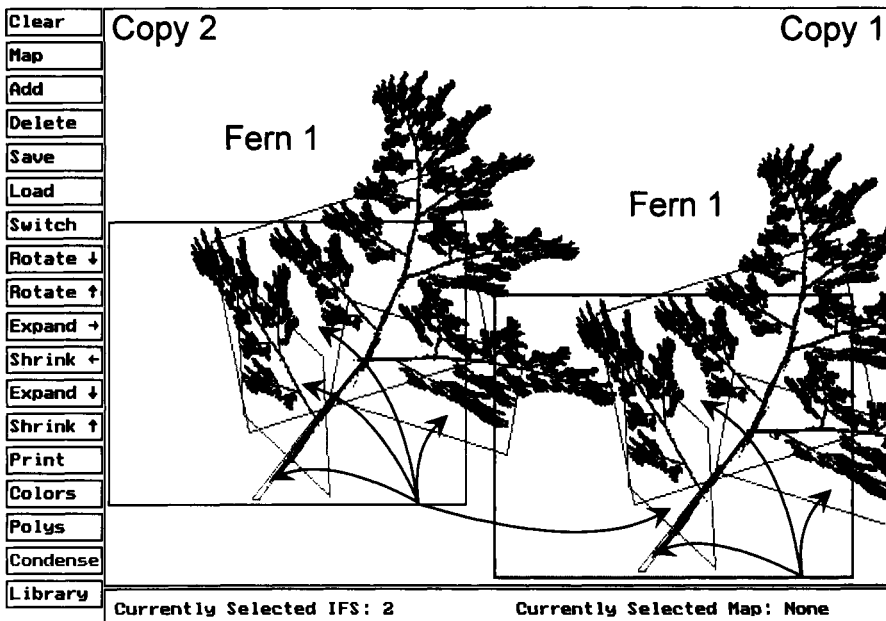


Figure X.278. This shows how the attractor in Figure X.277 is also the attractor of a recurrent fractal system, involving two copies of $\square$ instead of one. The attractor of the IFS appears as *part of* the attractor of a recurrent IFS.

Figure X.279. The IFS acting on the first copy of $\square$ is modified to create a new fern, Fern 2.

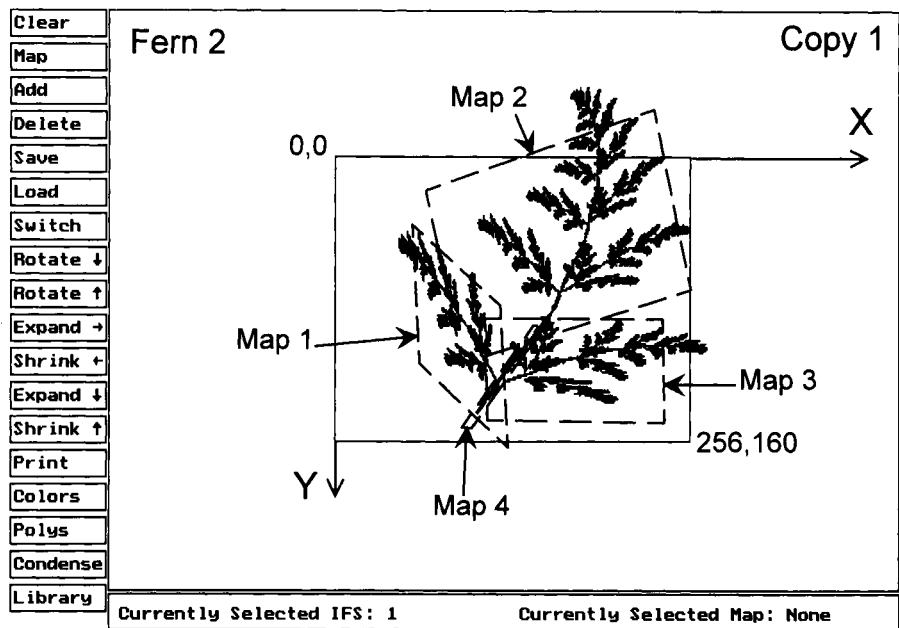


Figure X.280. The result of the modification of the first copy of the fern is to fundamentally alter the structure of the part of the attractor appearing on the first copy of $\square$.

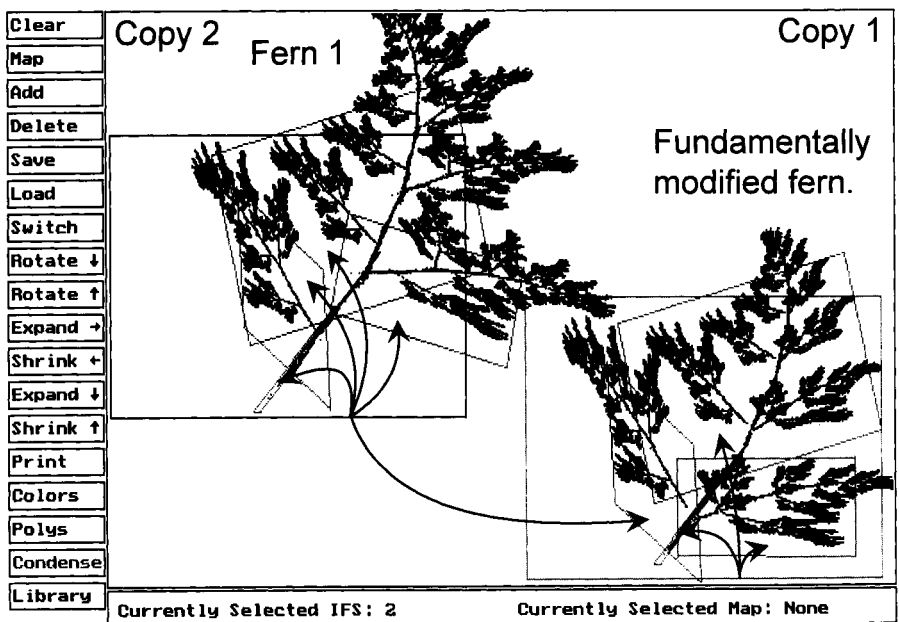


Table X.4. IFS code for Fern 1.

w	a	b	c	d	e	f
1	-0.02	0.39	-0.31	0.30	96	108
2	0.68	0.17	-0.17	0.68	73	11
3	0.05	-0.80	-0.22	-0.19	286	131
4	0.02	-0.36	0.00	0.38	160	97

Table X.5. IFS code for Fern 2.

w	a	b	c	d	e	f
1	-0.02	0.39	-0.31	0.30	64	116
2	0.64	0.16	-0.16	0.64	67	15
3	0.00	-0.80	-0.22	0.01	237	147
4	0.20	-0.33	0.00	0.35	144	96

smaller ferns with alternating fronds. We are learning to control a recurrent fractal model!

Finally what happens if we feed the first copy back into some of the fronds on the second copy? The results of such an adjustment are shown in Figure X.281. In fact, in this example, the bottom-most fronds on each fern are affine transformations of the other fern.

Color Plate 36 shows a fern image, and zooms on it, produced by a fractal system. In this case the fractal system involves four copies of $\square$ and a recurrent structure of affine transformations of the type discussed in this section. In this section each of three successive zooms on one of the ferns reveals new structure.

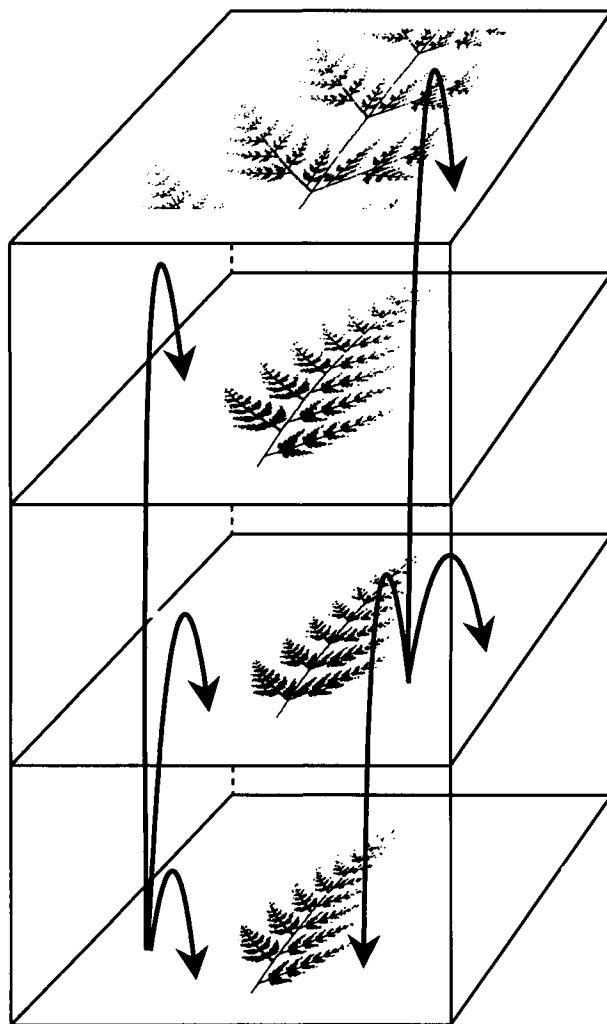
Another example of an attractor, consisting of four point-sets at once, is shown in Figure X.282. It is good to think of the vector of sets as a single entity.

4 Fractal Systems with Vectors of Measures as Their Attractors

In this section we are concerned with the construction of fractal systems, using recurrent IFS, whose attractors are vectors of measures. Our goal is to show how the collage theorem and associated machinery can be set up to enable fractal modelling using the measure theoretic attractors of certain recurrent IFS. That is, recurrent IFS can be used to make fractal systems suitable for modelling measures.

Recall what happens in the case of an IFS with probabilities: for example, suppose $\mathbb{X} = \square \subset \mathbb{R}^2$, $\{\mathbb{X}; w_1, \dots, w_N; p_1, \dots, p_N\}$ is an IFS with probabilities, as in section 6 in Chapter IX, with the w_i 's contractive affine, and $\mathbb{Y} = \mathcal{P}(\mathbb{X})$; then we can define a corresponding fractal system by taking $\mathcal{O}$ to be the associated Markov

Figure X.281. A fractal system, based on a recurrent IFS structure, produces two ferns whose structures are woven together. Neither could exist without the other!



operator as defined in section 6 in Chapter IX; that is $O : \mathbb{Y} \rightarrow \mathbb{Y}$ is defined by

$$O(v) = M(v) = p_1 v \circ w_1^{-1} + p_2 v \circ w_2^{-1} + \cdots + p_N v \circ w_N^{-1}.$$

Then we know from the theory of Chapter IX that O is contractive in the Hutchinson metric.

What is the analogous structure in the case of recurrent IFS? We use the notation of the previous section, but restrict attention to the case of measures supported on three copies of $\mathbb{X}$. Instead of using the measure space $\mathcal{P}(\mathbb{X})$, we work with the space of normalized Borel measures $\tilde{\mathcal{P}}$ that are defined on Borel subsets $\tilde{S}$ of $\mathbb{X} \times \mathbb{X} \times \mathbb{X}$ of the special form

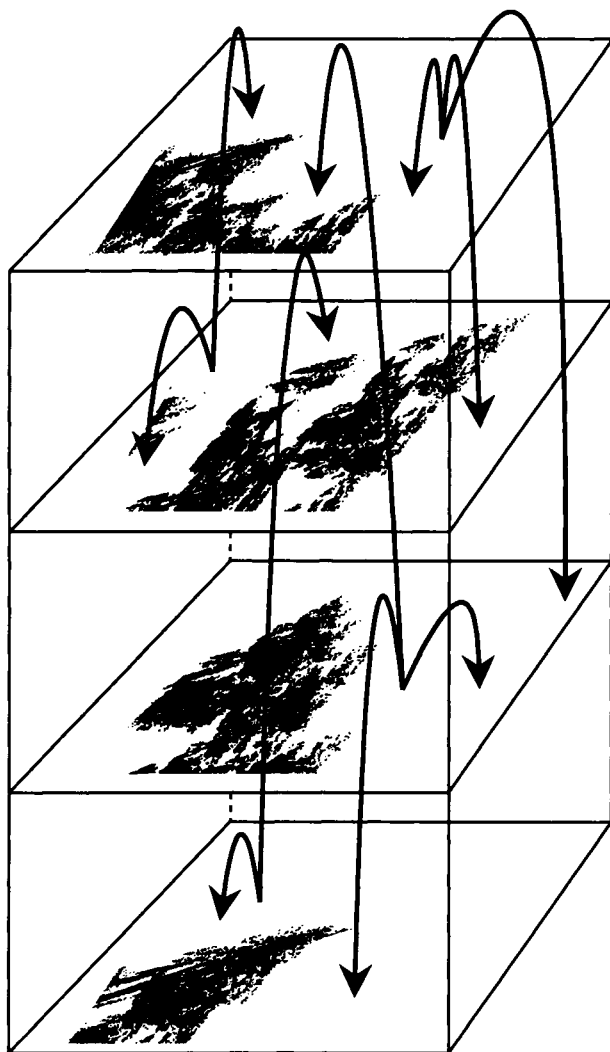


Figure X.282. Affine maps, linking four copies of $\square$, lead to a recurrent structure consisting of four linked sets. Adjustments of the coefficients lead to a wide variety of fractal modes in this fractal system.

$$\tilde{S} = (S_1, S_2, S_3) \in \mathcal{B}(\mathbb{X}) \times \mathcal{B}(\mathbb{X}) \times \mathcal{B}(\mathbb{X}).$$

where $\mathcal{B}(\mathbb{X})$ denotes the Borel subsets of $\mathbb{X}$. A point in $\tilde{\mathcal{P}}$ is described by a vector of measures

$$\tilde{\mu} = (\mu_1, \mu_2, \mu_3),$$

where each μ_i is a Borel measure on $\mathbb{X}$, and

$$\mu_1(\mathbb{X}) + \mu_2(\mathbb{X}) + \mu_3(\mathbb{X}) = 1.$$

Then the measure of vector of subsets $\tilde{S} = (S_1, S_2, S_3)$ is

$$\tilde{\mu}(\tilde{S}) = \mu_1(S_1) + \mu_2(S_2) + \mu_3(S_3).$$

We define a generalized Hutchinson metric $\tilde{d}$ on $\tilde{\mathcal{P}}$ by

$$\tilde{d}(\tilde{\mu}, \tilde{\nu}) = \sup \left\{ \sum_i \left(\int_{\mathbb{X}} f_i d\mu_i - \int_{\mathbb{X}} f_i d\nu_i \right) : \tilde{f} = (f_1, f_2, f_3) \in L \right\},$$

for all $\tilde{\mu}, \tilde{\nu} \in \tilde{\mathcal{P}}$, where

$$L = \left\{ \tilde{f} = (f_1, f_2, f_3) : f_i : \mathbb{X} \rightarrow \mathbb{R} \text{ continuous,} \right. \\ \left. \text{and } |f_i(x) - f_i(y)| \leq d(x, y) \quad \forall x, y \in \mathbb{X}, i = 1, 2, 3 \right\}.$$

Then, from the fact that $(\mathcal{P}(\mathbb{X}), d_H)$ is a compact metric space whenever $(\mathbb{X}, d)$ is a compact metric space, it follows that $(\tilde{\mathcal{P}}, \tilde{d})$ is a compact metric space too.

We are now ready to describe generalized Markov operators $\tilde{M}$,

$$\tilde{M} : \tilde{\mathcal{P}} \rightarrow \tilde{\mathcal{P}},$$

that are associated with recurrent IFS. We illustrate the idea for recurrent IFS such as the one in Table X.1, which is the subject of Figures X.268 and X.274. We simply define

$$\tilde{M}(\tilde{\mu}(\tilde{S})) = \left(\sum_i p_{i1} \mu_i(w_1^{-1}(S_1)), \sum_i p_{i2} \mu_i(w_2^{-1}(S_2)), \sum_i p_{i3} \mu_i(w_3^{-1}(S_3)) \right)$$

Our general expectation is that such operators are contractions on $(\tilde{\mathcal{P}}, \tilde{d})$ when the transformations w_i are contractions.

We demonstrate that $\tilde{M}$ has contractivity factor s when $p_{ij} = p_j$, and the w_i 's are contractive with contractivity factor $0 \leq s < 1$. We will need the identity

$$\int_{\mathbb{X}} f_j d(\tilde{M}(\tilde{\nu})_j) = \sum_i p_{ij} \int_{\mathbb{X}} f_j(w_j(x)) d\nu_i(x),$$

which follows from a change of variables. Then

$$\begin{aligned} & \tilde{d}(\tilde{M}(\tilde{\nu}), \tilde{M}(\tilde{\mu})) \\ &= \sup \left\{ \sum_j \left(\int_{\mathbb{X}} f_j d(\tilde{M}(\tilde{\mu})_j) - \int_{\mathbb{X}} f_j d(\tilde{M}(\tilde{\nu})_j) \right) : \tilde{f} \in L \right\}, \\ &= \sup \left\{ \int_{\mathbb{X}} \sum_{ij} p_{ij} f_j(w_j(x)) d\mu_i - \int_{\mathbb{X}} \sum_{ij} p_{ij} f_j(w_j(x)) d\nu_i : \tilde{f} \in L \right\}, \\ &= \sup \left\{ \int_{\mathbb{X}} \sum_{ij} p_j f_j(w_j(x)) d\mu_i - \int_{\mathbb{X}} \sum_{ij} p_j f_j(w_j(x)) d\nu_i : \tilde{f} \in L \right\}. \end{aligned}$$

Let, for $x \in \mathbb{X}$,

$$\hat{f}(x) = \left(s^{-1} \sum_j p_j f_j(w_j(x)), s^{-1} \sum_j p_j f_j(w_j(x)), s^{-1} \sum_j p_j f_j(w_j(x)) \right).$$

Then $\hat{f} \in L$. Let

$$\hat{L} = \{\tilde{f} \in L : \tilde{f} \text{ is of the special form of } \hat{f} \text{ above, for some } f \in L\}.$$

Then we can write

$$\tilde{d}(\tilde{M}(\tilde{v}), \tilde{M}(\tilde{\mu})) = \sup \left\{ s \int \tilde{f} d\tilde{\mu} - s \int \tilde{f} d\tilde{v} : \tilde{f} \in \tilde{L} \right\}.$$

Since $\hat{L} \subset L$, it follows that

$$\tilde{d}(\tilde{M}(\tilde{v}), \tilde{M}(\tilde{\mu})) \leq s \tilde{d}(\tilde{v}, \tilde{\mu}).$$

This completes the demonstration of the contractivity of the recurrent “multilevel” Markov operator, in the special case $p_{ij} = p_j$.

However, in the general case where the transition matrix $P = (p_{ij})$ is not constrained as above, the contractivity of the Markov operator is controlled not only by the contractivities of the w_i 's but also by the behavior of the transition matrix. To understand this, consider what happens when the space $\mathbb{X}$ consists of a single point. Then $\tilde{\mathcal{P}}$ becomes the set

$$\{\tilde{\mu} = (\mu_1, \mu_2, \mu_3) \in \mathbb{R}^3 : \mu_1 + \mu_2 + \mu_3 = 1, \mu_1 \geq 0, \mu_2 \geq 0, \mu_3 \geq 0\}.$$

That is, a typical measure $\tilde{\mu} \in \tilde{\mathcal{P}}$ is described by a vector of length three and total mass one. The Hutchinson distance between two such vectors is simply the sum of the absolute values of the differences between the components as follows:

$$\tilde{d}(\tilde{\mu} = (\mu_1, \mu_2, \mu_3), \tilde{v} = (v_1, v_2, v_3)) = |\mu_1 - v_1| + |\mu_2 - v_2| + |\mu_3 - v_3|.$$

Then the Markov operator $\tilde{M} : \tilde{\mathcal{P}} \rightarrow \tilde{\mathcal{P}}$ is given by

$$\begin{aligned} \tilde{M}(\tilde{\mu}) = (\mu_1, \mu_2, \mu_3)P &= (\mu_1 p_{11} + \mu_2 p_{21} + \mu_3 p_{31}, \mu_1 p_{12} + \mu_2 p_{22} + \mu_3 p_{32}, \\ &\mu_1 p_{13} + \mu_2 p_{23} + \mu_3 p_{33}), \end{aligned}$$

where

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix}.$$

It follows that

$$\tilde{d}(\tilde{M}(\tilde{v}), \tilde{M}(\tilde{\mu})) \leq \tilde{d}(\tilde{v}, \tilde{\mu}).$$

However we cannot, in the general case, assert a contractivity factor smaller than one, as can be seen by considering the case

$$P = \begin{bmatrix} 1 - 2\epsilon & \epsilon & \epsilon \\ \epsilon & 1 - 2\epsilon & \epsilon \\ \epsilon & \epsilon & 1 - 2\epsilon \end{bmatrix},$$

where ϵ is very small and positive. For this transition matrix, we find that

$$\tilde{d}(\tilde{M}((1, 0, 0)), \tilde{M}((0, 0, 1))) = (1 - 3\epsilon)\tilde{d}((1, 0, 0), (0, 0, 1)).$$

However, so long as the Markov operator has a unique fixed point, there always exist metrics such that the operator is contractive, in fact, with an arbitrarily small contractivity factor. The problem is the identification of such a metric, one that is simultaneously intuitively or visually accessible.

The operator $M : \mathcal{P}(\mathbb{X}) \rightarrow \mathcal{P}(\mathbb{X})$, associated with an IFS with probabilities,

$$M(\nu) = p_1\nu \circ w_1^{-1} + p_2\nu \circ w_2^{-1} + \cdots + p_N\nu \circ w_N^{-1},$$

is an example of a measure preserving transformation on a metric space, constructed from contractive transformations.

Given a set of spaces, $\{\mathbb{X}_k : k = 1, 2, \dots, N\}$, we can construct measure preserving transformations $M_{ij} : \mathcal{P}(\mathbb{X}_i) \rightarrow \mathcal{P}(\mathbb{X}_j)$, according to

$$M_{ij}(\nu) = p_{ij1}\nu \circ w_{ij1}^{-1} + p_{ij2}\nu \circ w_{ij2}^{-1} + \cdots + p_{ijN_{ij}}\nu \circ w_{N_{ij}}^{-1},$$

where $w_{ijn} : \mathbb{X}_i \rightarrow \mathbb{X}_j$ for $n \in \{1, 2, \dots, N_{ij}\}$ is contractive between the metric spaces $(\mathbb{X}_i, d_i)$ and $(\mathbb{X}_j, d_j)$ according to

$$d_j(w_{ijn}(x), w_{ijn}(y)) \leq s \cdot d_i(x, y) \quad \text{for all } x, y \in \mathbb{X}_i,$$

and the p_{ijn} 's being positive numbers such that

$$\sum_n p_{ijn} = 1.$$

(One can think of $\{\mathbb{X}_i, \mathbb{X}_j; w_{ijn}, p_{ijn}, n = 1, 2, \dots, N_{ij}\}$ as being an “IFS” that acts between the spaces $\mathbb{X}_i$ and $\mathbb{X}_j$.) We now introduce an $N \times N$ transition matrix $P = (p_{ij})$, that is row stochastic, and gives the probability of transition from space $\mathbb{X}_i$ to $\mathbb{X}_j$. Then we can define a Markov operator $\tilde{M} : \tilde{\mathcal{P}} \rightarrow \tilde{\mathcal{P}}$, where $\tilde{\mathcal{P}} = \mathcal{P}(\mathbb{X}_1) \times \mathcal{P}(\mathbb{X}_2) \times \cdots \times \mathcal{P}(\mathbb{X}_N)$, by

$$\tilde{M}(\mu_1, \mu_2, \dots, \mu_N) = \left(\sum_i p_{i1}M_{i1}(\mu_i), \sum_i p_{i2}M_{i2}(\mu_i), \dots, \sum_i p_{iN}M_{iN}(\mu_i) \right).$$

In general we expect such operators to be contractive, and to provide the basis for useful fractal systems for modelling vectors of measures, and in particular, grayscale photographs. An example of such a system is VRIFS⁵, which is an interactive image modelling system that uses a fractal system built from operators of the type of $\tilde{M}$

⁵ VRIFSTM, which stands for Vector Recurrent Iterated Function System, is an interactive measure theoretic fractal system for modelling images in two dimensions. It runs on Sun workstations and uses four-dimensional vectors of measures under affine control. It is available from Iterated Systems Inc., Commercial Division, 5550-A Peachtree Parkway, Norcross, Georgia, 30092, USA, telephone number (404) 840 0633.

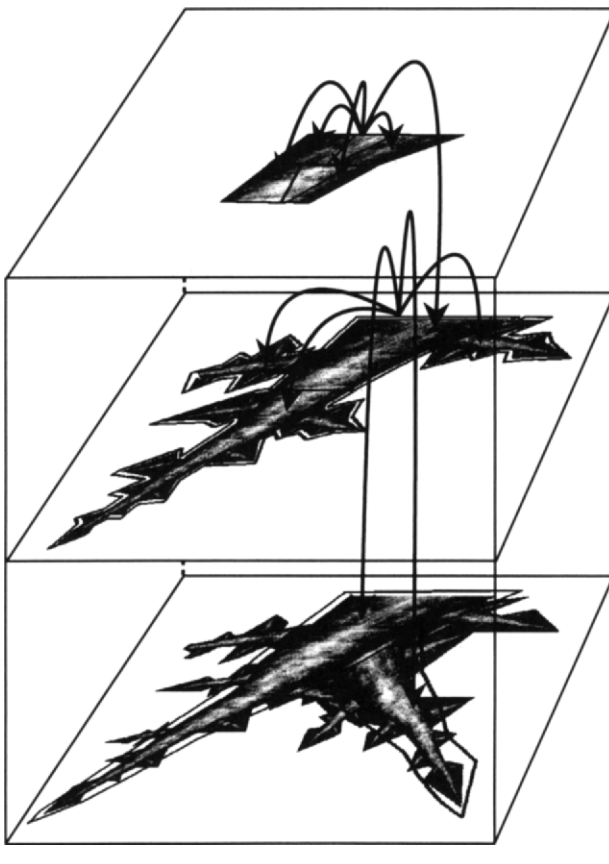


Figure X.283. A special Markov operator acts on a vector of measures. Its attractor is a stack of images, including a split root. Fractal systems of this type are good for fractal modelling of grayscale images.

above, with all transformations being two dimensional affine transformations. An example of an image generated by VRIFS™ is shown in Figure X.283. See also Color Plates 37, 38, and 39.

5 References

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